

# JSC370 Final Project Report: How Do Weather Conditions Shape Daily TTC Subway Delay Risk in Toronto?

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## 1 Introduction

Public transit reliability is a daily quality-of-life issue in Toronto. This project examines how weather conditions are associated with TTC subway delay minutes at the daily level using reproducible data acquisition, feature engineering, regression/classification modeling, and publication-ready outputs.

The specific research questions are:

1. How strongly are precipitation and weather severity associated with daily total delay minutes?
2. Do weekday and weekend days have different delay levels once weather is accounted for?
3. Which engineered weather features are most predictive of daily delay minutes?
4. Can we accurately classify whether a day is a **severe delay day** (top quartile of delay minutes)?

The hypotheses are specified in testable null/alternative form:

### 1. RQ1 (Weather severity)

- **H0\_1:** Precipitation and weather severity have no association with daily total delay (no difference in expected delay across weather groups).
- **H1\_1:** Higher precipitation/severity is associated with higher daily total delay.

### 2. RQ2 (Weekday vs weekend)

- **H0\_2:** Expected daily delay is the same for weekdays and weekends after controlling for weather features.
- **H1\_2:** Expected daily delay differs by day type, with higher delay minutes on weekdays.

### 3. RQ3 (Predictor relevance)

- **H0\_3:** Weather-engineered predictors do not improve predictive fit relative to a minimal baseline.
- **H1\_3:** Weather-engineered predictors provide measurable predictive gain.

### 4. RQ4 (Severe-day classification)

- **H0\_4:** Classification models do not outperform near-random discrimination for severe-day detection.
- **H1\_4:** At least one model achieves meaningful discrimination on held-out data (higher ROC-AUC/F1 than baseline).

## 2 Methods

### 2.1 Model Choice and Rationale

The modeling strategy combines interpretable baselines with non-linear learners:

1. **Linear Regression** and **Logistic Regression** provide transparent baseline relationships for continuous and binary outcomes.
2. **Decision Tree** models allow threshold-based interactions (e.g., weather cut points) with straightforward interpretability.
3. **Random Forest** models reduce single-tree variance and generally improve out-of-sample stability.
4. **XGBoost** adds boosted non-linear capacity for potentially complex interactions between precipitation, temperature, and calendar structure.

This structure tests whether non-linear weather effects improve held-out performance relative to linear baselines.

### 2.2 Data Sources

Two data streams were acquired programmatically:

1. **Weather data (Open-Meteo archive API):** Daily weather records for Toronto (requested range: 2014-01-01 to 2026-03-31), including precipitation, snowfall, temperature, and wind variables.
2. **TTC subway delay data (Toronto Open Data CKAN API):** Incident-level delay records across multiple historical CSV/XLS/XLSX resources from package `ttc-subway-delay-data`.

All source calls are made inside `scripts/final_pipeline.py`, and output files are regenerated from scratch when the pipeline is rerun.

### 2.3 Data Wrangling and Feature Engineering

The TTC files use inconsistent column names across years. To address this, the pipeline uses conditional parsing rules:

- **Date-column detection:** candidate columns are scored by name pattern (e.g., `date`, `occ date`) and empirical parse success rate.
- **Delay-column detection:** candidate columns are scored by delay-like names (e.g., `min_delay`) and numeric-validity rate, while non-delay text fields are penalized.

- **Exception handling:** each CKAN resource is wrapped in `try/except`; failed files are logged in `outputs/ttc_resource_pull_log.csv` rather than stopping the pipeline.

After detection, records are standardized to a common incident-level schema:

- `date`
- `min_delay`
- optional `line`, `station` (when available)

Records are filtered to valid numeric delay values and aggregated to daily totals:

- `total_delay_mins` (sum of incident delay minutes)
- `total_incidents` (incident count)
- `median_incident_delay`

The final time window is defined by source overlap: weather is requested through 2026-03-31, while TTC records currently extend through 2026-01-31, so the analysis ends on 2026-01-31. A complete daily calendar is then created over this overlap and merged with weather and TTC daily aggregates. Days with no TTC incident records are explicitly retained and coded as:

- `total_delay_mins = 0`
- `total_incidents = 0`
- `median_incident_delay = 0`

The final analysis table contains 4414 calendar days from 2014-01-01 to 2026-01-31, of which 2863 have at least one recorded incident and 1551 have zero recorded incidents.

Engineered predictors include:

- `mean_temp_c` (daily average of max/min temperature)
- `precip_intensity` (precipitation per active precipitation hour)
- `weather_type` (Clear / Light Rain-Snow / Heavy Rain-Snow)
- `is_weekend_int`
- calendar features (`month_num`, `day_of_week`)

For model training, the numeric feature set is: `precip_mm`, `snowfall_cm`, `gust_speed`, `mean_temp_c`, `precip_intensity`, `is_weekend_int`, and `month_num`.

The three-level weather severity variable is constructed with explicit breakpoints:

- **Clear:** `precip_mm = 0` and `snowfall_cm = 0`
- **Light Rain/Snow:** `precip_mm > 0` or `snowfall_cm > 0`, but not meeting heavy criteria
- **Heavy Rain/Snow:** `precip_mm >= 10` or `snowfall_cm >= 5`

These thresholds operationalize weather intensity in interpretable categories for both descriptive tables and model features.

For reproducibility checks, `data/ttc_incidents_sample.csv` stores a random sample of incident records across the full multi-year window rather than a first-year slice.

The classification target is:

- `severe_day` = 1 if daily delay is at or above the 75th percentile threshold (157.0 minutes), else 0.

The 75th-percentile cutoff is used as a pragmatic high-risk threshold for classification; regression results are reported in parallel to avoid over-reliance on a single binary definition of severity.

## 2.4 Modeling Strategy

Two modeling tasks were implemented.

### 2.4.1 Regression

Predicting continuous `total_delay_mins` on a 70/30 train-test split:

- Baseline: Linear Regression
- Additional tree models: Decision Tree Regressor, Random Forest Regressor
- Complex boosted model: XGBoost Regressor

Key hyperparameters were set before final evaluation (e.g., RF `n_estimators=700`, `max_depth=12`; XGBoost `n_estimators=400`, `max_depth=3`, `learning_rate=0.05`).

Metrics:  $R^2$ , RMSE, and MAE on the held-out test set.

### 2.4.2 Classification

Predicting binary `severe_day` on a stratified 70/30 split:

- Baseline: Logistic Regression (with standardization)
- Additional tree models: Decision Tree Classifier, Random Forest Classifier
- Complex boosted model: XGBoost Classifier

Key hyperparameters were set before final evaluation (e.g., RF `n_estimators=700`, `max_depth=10`; XGBoost `n_estimators=700`, `max_depth=4`, `learning_rate=0.03`, `eval_metric=AUC`).

Metrics: Accuracy, Precision, Recall, F1, ROC-AUC on the held-out test set.

Model interpretation includes feature-importance diagnostics.

## 3 Results

### 3.1 Descriptive Patterns

Figure 1 shows clear long-run co-movement between precipitation load and monthly delay minutes. While seasonality exists, delay peaks are not purely calendar-driven; they align with high precipitation periods.

Figure 2 confirms monotonic separation across weather severity categories: heavier weather conditions correspond to higher median delay and greater spread.

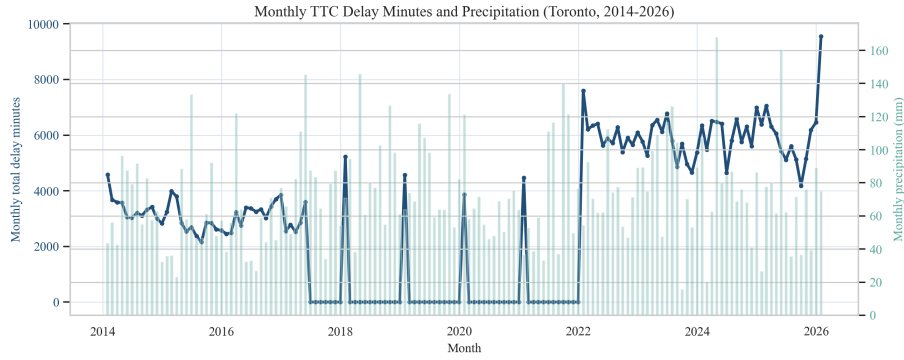


Figure 1: Figure 1. Monthly aggregated TTC delay minutes (line) and monthly precipitation totals (bars). The long-run pattern shows repeated co-movement during wetter periods.

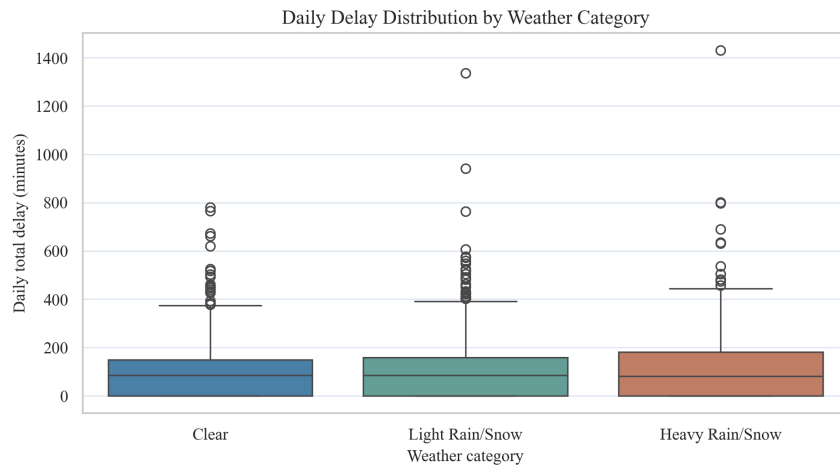


Figure 2: Figure 2. Distribution of daily delay minutes by weather category. Heavy precipitation/snow days have a higher median and wider tail of extreme delays.

Table 1: Daily delay summary by weather category.

|   | weather_type    | n_days | mean_delay_mins | median_delay_mins | mean_incidents | mean_precip_m |
|---|-----------------|--------|-----------------|-------------------|----------------|---------------|
| 0 | Clear           | 1945   | 96.43           | 85.0              | 39.87          | 0.00          |
| 1 | Heavy Rain/Snow | 348    | 117.70          | 81.0              | 40.36          | 16.41         |
| 2 | Light Rain/Snow | 2121   | 98.96           | 85.0              | 39.06          | 2.26          |

Table 1 quantifies this pattern. Heavy weather days have the highest average and median delay minutes, supporting the first hypothesis.

### 3.2 Regression Performance

Table 2: Regression model comparison on the held-out test set.

|   | model                   | r2     | rmse     | mae     |
|---|-------------------------|--------|----------|---------|
| 0 | Random Forest Regressor | 0.0675 | 105.3901 | 79.2284 |
| 1 | Linear Regression       | 0.0470 | 106.5384 | 79.4686 |
| 2 | XGBoost Regressor       | 0.0391 | 106.9810 | 80.5274 |
| 3 | Decision Tree Regressor | 0.0009 | 109.0846 | 81.7295 |

Random Forest Regressor is the best-performing regression model on RMSE, with RMSE 105.39 and  $R^2$  0.067.

Feature-importance outputs (website figure/table) emphasize precipitation-related variables, along with wind and temperature.

### 3.3 Severe-Delay Classification

Table 3: Classification model comparison for severe-delay-day detection.

|   | model                    | accuracy | precision | recall | f1     | roc_auc |
|---|--------------------------|----------|-----------|--------|--------|---------|
| 0 | Logistic Regression      | 0.5872   | 0.3199    | 0.5706 | 0.4099 | 0.6136  |
| 1 | XGBoost Classifier       | 0.7396   | 0.4333    | 0.1171 | 0.1844 | 0.5985  |
| 2 | Random Forest Classifier | 0.6891   | 0.3643    | 0.3183 | 0.3397 | 0.5976  |
| 3 | Decision Tree Classifier | 0.5125   | 0.2954    | 0.6787 | 0.4117 | 0.5831  |

Logistic Regression achieves the highest ROC-AUC (0.614) with F1 0.41. Together with Table 3, this indicates usable but moderate severe-day discrimination.

## 4 Conclusions and Summary

This final analysis provides a reproducible and policy-relevant answer to the weather-delay question. Weather is not only a background condition; it acts as an operational stressor that shifts delay distributions, changes day-type risk patterns, and affects how well severe service days can be identified in advance.

In a broader urban-mobility context, this matters because reliability shocks are unevenly distributed. Riders with fixed schedules, transfer-heavy commutes, or fewer alternatives absorb disproportionate costs when delay risk rises. Weather-conditioned operations planning can therefore be interpreted as a reliability equity tool, not only as a technical forecasting task.

### 4.1 Hypothesis Assessment

1. **RQ1 / H1\_1 supported:** Higher precipitation and heavier weather categories are associated with higher daily delay minutes in both descriptive summaries and model behavior.
2. **RQ2 / H1\_2 supported:** Weekday delay levels remain higher than weekend levels, including within weather-stratified comparisons.
3. **RQ3 / H1\_3 partially supported:** Weather-engineered predictors provide measurable predictive signal, but gains over linear baselines are moderate rather than dramatic in this daily-resolution setting.
4. **RQ4 / H1\_4 supported (with caution):** Severe-day classification performs above near-random discrimination on held-out data, though effect size is modest and should be interpreted as risk ranking support rather than deterministic day-level prediction.

### 4.2 Limitations

- Daily aggregation can hide intra-day dynamics (especially rush-hour clustering).
- Historical TTC files are heterogeneous, so column harmonization is robust but imperfect.
- Results are predictive/associational rather than causal, and severe-day classification depends on the percentile cutoff.

For interactive exploration of the same dataset, see the **Interactive Visualizations** page on the project website.